

Alexander Brown

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Skills

Technical: SolidWorks, Onshape, MATLAB, GD&T (ASME Y14.5), AS9102 FAI, ERP, Microsoft Office Suite, FEA, Python, C++

Professional: Shop-floor problem solving, cross-functional communication, technical documentation, adaptability

Summary

U.S. Navy veteran and Mechanical Engineering student with aerospace manufacturing experience in configuration control, inspection planning, AS9102 FAI support, nonconformance tracking, ERP workflows, and production process improvement. Hands-on hardware focus spanning DFM, tooling, work instructions, and scaling defense systems. Available full time in 2027.

Experience

Manufacturing Quality Engineering Intern, Verus Aerospace – Tacoma, WA Dec 2025 – Present

- Supported rapid production of mission-critical aerospace hardware by maintaining Engineering Masters, revision control, and ERP tracking across thousands of active components, improving configuration traceability and release readiness
- Supported recovery of a delayed supplier production issue involving 10 backlogged orders and approximately 40 parts by assisting with a supplier audit, identifying scrap and process-control concerns, and improving corrective action visibility between quality, manufacturing, and supplier stakeholders
- Improved quality verification before hardware release by creating inspection plans, supporting AS9102 First Article Inspection activities, and performing independent over-check inspections on aerospace assemblies and major OEM hardware to identify defects, dimensional discrepancies, and documentation gaps
- Led Quality Clinic operations for non-conforming hardware by tracking parts through rework, reassignment, and scrap disposition while redesigning material flow, visual organization, and disposition processes to improve traceability, communication, and turnaround efficiency

Multi-Tool Fabrication Project, University of Washington – Tacoma, WA May 2026 – June 2026

- Fabricated a functional folding multi-tool assembly using manual mill and CNC machining operations, manufacturing mating components including arms, jaws, pivot features, and bottle opener geometry
- Interpreted engineering drawings and machined aluminum components to tolerance requirements including ± 0.010 in general dimensions, ± 0.005 in precision dimensions, 0.003 in parallelism, and 0.002 in flatness controls
- Calculated feeds, speeds, and machining parameters for aluminum to achieve required surface finish, dimensional accuracy, and repeatable material removal across multiple operations
- Verified part quality using calipers, gauges, and dial indicators, recording measurements against drawing tolerances to confirm assembly fit and identify dimensional variation

Gearbox Project, University of Washington – Tacoma, WA May 2026 – June 2026

- Designed a shaft-reduction gearbox for a 1-degree-of-freedom robotic elbow by coordinating CAD development, mechanical calculations, and design review tasks under a fixed project deadline
- Developed a fully constrained CAD assembly to validate component fit, shaft alignment, gear placement, motion clearance, bearing support, and system integration before final design submission
- Verified gearbox performance requirements by performing gear ratio selection, torque transmission, shaft sizing, bearing selection, and load calculations for the robotic joint
- Improved manufacturability and assembly repeatability by applying GD&T principles, tolerance stack-up analysis, and clearance/interference fit checks to reduce alignment risk and control backlash

Hospital Corpsman, U.S. Navy – Various Locations Aug 2018 – May 2023

- Managed a \$102K medical inventory system across 6,200+ items, reducing supply bottlenecks and supporting 20% lower operational downtime in a fast-paced military environment
- Enforced strict safety engineering protocols for 50+ hazardous materials by creating comprehensive SDS documentation to achieve 100% federal regulatory compliance

Education

University of Washington – BS in Mechanical Engineering June 2027